ZIFENG WEN

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EDUCATION

Imperial College London *Sep. 2026-Sep. 2027*
MSc in Advanced Computing *London, United Kingdom*

- **Core Modules:** Software Engineering for Industry; Software Engineering for Machine Learning Systems, Machine Learning; Generative AI; Natural Language Processing; Machine Learning for Imaging; Computational Optimisation; Scheduling and Resource Allocation.

Harbin Institute of Technology, Shenzhen (Project 985) *Aug. 2022-Jul. 2026*
BEng in Computer Science and Technology *Shenzhen, China*

- **GPA:** 94.21/100 (3.93 / 4.00) **RANK:** 2 / 317 **(Top 1%)**
- **Core Modules:** Large Language Models: Foundations, Practice and Challenges (100), Mathematical Logic(100), ACM Programming(100), Artificial Intelligence(98), Introduction to Object-oriented Software Construction(98), Formal Language and Automata Theory(97)
- **Outstanding Undergraduate Thesis Award**, School of Computer Science and Technology (**Top 16** Recipients in the Class of 2026)

Imperial College London *Jan. 2024-Feb. 2024*
2024 Winter Data Science Project *London, United Kingdom*

- Developed a 3D brain tumour segmentation system based on multi-modal MRI scans (T1, T1c, T2 and FLAIR), using a **U-Net backbone** to generate tumour masks for accurate segmentation and visual analysis.
- Achieved **Distinction** (Grade A), and was awarded the **Best Presentation Award** and **Second-Class Scholarship** for outstanding project performance.

PUBLICATIONS

Yiqing Huang, Yixuan Kong, **Zifeng Wen**, Jiaole Wang, Shuang Song*, *"Design of a Miniaturized Wire-Driven Continuum Surgical Robot for Remote Operation in Minimally Invasive Surgery"*, presented at the 2025 IEEE International Conference on Real-time Computing and Robotics **(IEEE RCAR 2025)**

INTERNSHIP EXPERIENCE

Xiaohongshu (RED), Business Technology Department, Community Engineering Team *Mar. 2026-Jun. 2026*
AI Product Engineer Intern (AI-powered Community Systems) *Shanghai, China*

- **Multimodal Comment Intent Classification & On-device Inference:** Developed a multimodal comment-intent prediction system by combining user behaviour, session signals, post semantics, and engagement features; trained XGBoost and LinearSVC models and deployed them to iOS Core ML for real-time on-device inference.
- **LLM-powered Subscription Agent:** Designed and implemented an LLM-powered agent that identifies users' follow-up intentions from comments (e.g., author updates, event progress, and product releases); built subscription-target extraction, content retrieval, and update-detection pipelines to enable automated subscriptions and personalised notifications.
- **Knowledge Organisation Agent:** Built an intelligent content-organisation system using embedding-based semantic understanding and clustering algorithms; designed a human-in-the-loop agent workflow that automatically generates, refines, and maintains thematic collections for long-term knowledge management.

Insta360, Technology Center, Application Software Department *Nov. 2025-Mar. 2026*
Android Software Engineer Intern *Shenzhen, China*

- **On-Device AI Productionisation:** Collaborated with research teams to deploy computer vision models into the Android media platform; designed inference integration interfaces, model-output processing pipelines, and downstream business-consumption workflows, bridging research prototypes and large-scale production deployment for gallery and asset-management scenarios.
- **Live Photo Engineering:** Integrated Live Photo SDKs from multiple mobile vendors and implemented end-to-end support for browsing, downloading, and exporting Live Photos within the Insta360 application.
- **Project Delivery:** Delivered 25 high-priority tasks, including 16 feature developments and 9 production issue fixes, with all contributions successfully released to the mainline Insta360 App.

Meituan, Core Local Commerce, Video Technology Department, Live Streaming Marketing Team *Apr. 2025-Aug. 2025*
Software Engineer Intern *Beijing, China*

- **Anchor Permission Governance:** Refactored the permission configuration workflow for live-streaming lottery campaigns; introduced cache- and API-based dual validation mechanisms to enable permission verification and automated access control.
- **Marketing Change Traceability:** Designed and implemented an AOP-based change tracking framework for core marketing workflows, including Red Packet Rain and viral sharing campaigns; enabled end-to-end modification auditing and efficient root-cause analysis without intrusive business-code changes.
- **SQL Architecture Optimisation:** Migrated embedded Java SQL statements to MyBatis XML mappings; introduced consistency validation and field-level monitoring mechanisms to improve maintainability, observability, and troubleshooting efficiency.
- **Project Delivery:** Delivered 14 high-priority projects, including 10 business features and 4 technical initiatives, all successfully deployed to production and supporting core live-streaming marketing scenarios.

ValueHD Corporation, Video Research Institute, AI Department *Dec. 2024-Feb. 2025*
Computer Vision Algorithm Engineer Intern *Shenzhen, China*

- Developed high-fidelity 3D human reconstruction pipelines based on **NeRF** and **3D Gaussian Splatting (3DGS)**, enabling animatable digital-human generation and rendering.
- Built and benchmarked NeRF- and 3DGS-based reconstruction systems; optimized rendering quality, training efficiency, memory

usage, and real-time performance through systematic architecture and parameter evaluation.

- Reproduced a diffusion-based human-scene interaction generation model on the **TRUMANS** dataset and integrated it with NeRF/3DGS rendering pipelines to generate temporally coherent human-scene interaction sequences.

RESEARCH EXPERIENCE

Agent Memory Systems for LLM Agents *Mar. 2026-Jul. 2026*

Research Assistant (Supervisor: Prof. Xuebo Liu, HITSZ)

Project Description: Investigated long-term memory mechanisms for LLM-based Software Engineering Agents to address challenges in historical experience utilisation and context window limitations. Conducted literature review and experimental studies on Agent Memory systems using SWE-bench as the evaluation benchmark.

- Conducted a systematic review of representative Agent Memory systems, including **AgeMem, G-Memory, MemGas, and BMAM**, analysing long-term memory, shared memory, and hierarchical memory architectures for LLM-based agents.
- Designed and implemented an Agent Memory experimental framework incorporating trajectory storage, semantic retrieval, dynamic memory updates, and multi-agent collaboration to support experience reuse and memory-augmented reasoning.
- Built an automated evaluation pipeline based on **SWE-bench**, benchmarking different memory strategies in terms of task success rate, inference cost, and context utilisation efficiency for Software Engineering Agents.

FoldScope: Immersive VR Platform for Dynamic Protein Structure Visualization and AI-assisted Design *Mar. 2025-Jul. 2025*

Researcher Assistant (Supervisor: Prof. Junjie Chen, HITSZ)

Project Description: Developed an immersive protein design and validation platform integrating diffusion-based protein generation models with structure-function evaluation workflows in a VR environment, enabling a closed-loop process from backbone generation and sequence design to structural verification.

- Developed a Python backend service to bridge the VR frontend (C#) and protein design models, parsing control parameters and PDB structures while integrating RFdiffusion for structure-constrained backbone generation.
- Integrated ProteinMPNN for structure-conditioned sequence design and RF3 for structure prediction and validation, with online evaluation using RMSD and pTM metrics.
- Built a closed-loop immersive VR platform combining protein generation, sequence design, structural verification, and visual analysis, resulting in **2 Chinese National Invention Patent Applications and 1 Software Copyright Registration.**

PERSONAL PROJECT EXPERIENCE

Learn-Claude-Code: Building an Agent Harness from First Principles *May. 2026-Jun. 2026*

- Reimplemented core Claude Code mechanisms including Agent Loop, Tool Use, Permission Control, Context Compaction, Skill Loading, Memory Management, Task Scheduling, and MCP-based capability routing.
- Designed modular agent infrastructure around a unified execution loop, supporting tool dispatch, subagent isolation, task dependency management, background execution, and long-context memory retrieval.
- Analysed and reproduced key agent-system design patterns, gaining hands-on experience in context engineering, agent orchestration, capability governance, and scalable AI-agent architecture.

HelloAgents: Multi-Agent Framework with Memory, Tool Use, and RAG *Feb. 2026-Mar. 2026*

- Built a lightweight multi-agent framework supporting ReAct, Plan-and-Solve, and Reflection paradigms; implemented a unified abstraction layer for LLM interaction, tool execution, and Thought→Action→Observation reasoning workflows.
- Designed a hierarchical memory system (Working, Episodic, Semantic, and Perceptual Memory) with a MemoryManager module; integrated Qdrant-based vector retrieval and RAG pipelines for long-term knowledge access.
- Developed an end-to-end travel-planning application using Vue3, FastAPI, and Amap MCP, featuring multi-agent orchestration, tool calling, itinerary generation, route planning, and structured output validation through Pydantic models.

MiniMind: End-to-end Tiny LLM Training and Inference Framework *Jan. 2026-Feb. 2026*

- Built a decoder-only Transformer framework (26M – 145M parameters) from scratch using native PyTorch, covering pretraining, supervised fine-tuning (SFT), knowledge distillation, and inference deployment.
- Implemented modern LLM architecture components including Pre-LN, RMSNorm, SwiGLU, RoPE with YaRN scaling, GQA, and KV Cache optimisation to support stable long-context training and efficient autoregressive inference.
- Developed distributed and mixed-precision training infrastructure with DDP, AMP, checkpoint recovery, and experiment management; integrated PyTorch 2 scaled_dot_product_attention kernels for efficient attention computation while preserving causal consistency.

AWARDS

Academic Awards:

- **Academic Undergraduate Scholarships**: Second-Class Academic Scholarship (2024 – 2025); Third-Class Academic Scholarship (2023 – 2024); Second-Class Academic Scholarship (2022 – 2023)

Competition Awards:

- Second Prize, Lanqiao Cup C/C++ Programming Competition (2023)
- Third Prize, 2023 China Undergraduate Mathematical Contest in Modeling (CUMCM)

LANGUAGES & TECHNICAL SKILLS

- **IELTS:** Overall 8.0 (L7.5, R8.5, W8.5, S6.5), **CEFR C1**
- Python, Java, MySQL | Agent Memory, MCP, RAG, Multi-Agent Systems,Prompt Engineering | PyTorch, FastAPI, Spring Boot